

# The Full-Dimensional Linear Cahn–Hilliard Spinodal Atlas

Route-A source-local certificate HCS-C304

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## Abstract

For every finite  $d \geq 1$ ,  $\kappa > 0$ , and  $\alpha \in \mathbb{R}$ , we classify the linear Cahn–Hilliard flow  $u_t = -\kappa \Delta^2 u - \alpha \Delta u$  on the  $2\pi$ -periodic torus. The result gives the self-adjoint analytic trace-class semigroup, exact lattice-shell spectrum, and the complete stable/critical/spinodal atlas. It also closes the energy law, Morse index and kernel, all fastest-shell ties, actual-support asymptotics, recurrence, and the singular  $\kappa = 0$  face. A 1653-cell archive with independent checker, symbolic, replay, mutation, and deterministic-build lanes is regression evidence; it is not the proof.

## 1 One theorem for the entire parameter space

Let  $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ ,  $H = L_0^2(\mathbb{T}^d)$ , and

$$A = -\kappa \Delta^2 - \alpha \Delta, \quad D(A) = H^4(\mathbb{T}^d) \cap H, \quad \kappa > 0, \quad \alpha \in \mathbb{R}.$$

Write  $r_d(n) = \#\{k \in \mathbb{Z}^d : |k|^2 = n\}$ .

**Theorem 1** (Full linear spinodal atlas). *For every finite  $d \geq 1$  the operator  $A$  is self-adjoint, bounded above, and has compact resolvent. It generates a self-adjoint analytic semigroup  $S(t)$ , trace class for every  $t > 0$ . On shell  $n > 0$ ,*

$$Ae_k = \sigma_n e_k, \quad \sigma_n = \alpha n - \kappa n^2, \quad (1)$$

*with multiplicity  $r_d(n)$ . The mean-zero equilibrium is exponentially stable iff  $\alpha < \kappa$ , critical iff  $\alpha = \kappa$ , and spinodally unstable iff  $\alpha > \kappa$ . Its unstable Morse index and kernel dimension are*

$$M = \sum_{n < \alpha/\kappa} r_d(n), \quad K = \sum_{n = \alpha/\kappa} r_d(n), \quad (2)$$

*where absent shells contribute zero. At criticality  $K = r_d(1) = 2d$ .*

*The spectral bound is attained precisely on*

$$\mathcal{N}_* = \operatorname{argmax}_{n \geq 1, r_d(n) > 0} (\alpha n - \kappa n^2), \quad (3)$$

*including every tie. For  $u_0 \neq 0$ , let  $\lambda_* = \max\{\sigma_n : P_n u_0 \neq 0\}$ . Then*

$$e^{-\lambda_* t} S(t) u_0 \longrightarrow \sum_{\sigma_n = \lambda_*} P_n u_0 \quad \text{in } L^2. \quad (4)$$

*Every recurrent state is stationary; hence there is no nonstationary periodic solution.*

*Proof.* For  $e_k = (2\pi)^{-d/2} e^{ik \cdot x}$ ,  $\Delta e_k = -|k|^2 e_k$ , so (1) follows. The real diagonal values tend to  $-\infty$  as  $|k| \rightarrow \infty$ , proving self-adjointness, compact resolvent, and an upper bound. The spectral theorem yields the analytic semigroup, while

$$\sum_{k \neq 0} e^{t(\alpha|k|^2 - \kappa|k|^4)} < \infty \quad (t > 0)$$

by quartic domination, proving trace class. The sign factorization  $\sigma_n = n(\alpha - \kappa n)$  proves the chamber and index formulas.

The maximization in (3) is over the whole represented spectrum, not a numerical cutoff. Indeed

$$\sigma_n = \frac{\alpha^2}{4\kappa} - \kappa \left( n - \frac{\alpha}{2\kappa} \right)^2. \quad (5)$$

If  $\alpha/\kappa \leq 1$ , then for  $n > 1$ ,  $\sigma_n - \sigma_1 = (n-1)[\alpha - \kappa(n+1)] < 0$ . If  $\alpha/\kappa > 1$ , shell one has positive rate whereas every  $n \geq \alpha/\kappa$  has nonpositive rate. Thus an explicit finite set of represented integers contains every maximizer, and all ties are retained. For a fixed datum the same decay at infinity gives a largest rate in its actual support. Factoring  $e^{\lambda_* t}$  and dominated convergence prove (4). Finally, recurrence along  $t_j \rightarrow \infty$  forces  $e^{t_j \sigma_n} \hat{u}_0(k) \rightarrow \hat{u}_0(k)$  coefficientwise; every nonzero coefficient must have  $\sigma_n = 0$ , so the state is stationary.  $\square$

## 2 Gradient-flow law and singular faces

Set

$$\mathcal{F}(u) = \frac{1}{2} \int_{\mathbb{T}^d} (\kappa |\nabla u|^2 - \alpha |u|^2) dx, \quad \mu = -\kappa \Delta u - \alpha u.$$

The equation is  $u_t = \Delta \mu$ . Periodic integration by parts gives the exact dissipation identity

$$\frac{d}{dt} \mathcal{F}(u(t)) = \langle \mu, \Delta \mu \rangle = -\|\nabla \mu\|_2^2. \quad (6)$$

This explains the energy signature without turning linear instability into a nonlinear saturation or coarsening theorem.

**Proposition 1** (Boundary completion). *On full  $L^2$ , the constant mode is stationary and equals the conserved mean. On the mean-zero space  $H$ , the singular face  $\kappa = 0$  is interpreted with its natural generator domain: for  $\alpha \neq 0$ , the operator  $-\alpha \Delta$  has domain  $H^2(\mathbb{T}^d) \cap H$ , while for  $\alpha = 0$  the zero generator has domain all of  $H$ . It gives forward heat when  $\alpha < 0$ , the identity when  $\alpha = 0$ , and no bounded  $L^2$   $C_0$  semigroup when  $\alpha > 0$ .*

*Proof.* The zero shell has rate zero. On the stated natural domains, the  $\kappa = 0$  positive shells have rates  $\alpha n$ . Negative  $\alpha$  gives heat, zero gives identity, and positive  $\alpha$  makes  $\sup_n e^{t \alpha n} = \infty$  for every  $t > 0$ .  $\square$

| Face                   | Exact conclusion                                                           |
|------------------------|----------------------------------------------------------------------------|
| $\alpha < \kappa$      | every mean-zero shell decays exponentially.                                |
| $\alpha = \kappa$      | only shell one is neutral; its dimension is $2d$ .                         |
| $\alpha > \kappa$      | finitely many represented shells are unstable.                             |
| fastest tie            | retain the entire tied spectral projection.                                |
| missing ambient leader | use the largest rate in the actual support of $u_0$ .                      |
| $\kappa \downarrow 0$  | heat / identity / no bounded semigroup according to the sign of $\alpha$ . |

## 3 Finite evidence, hostile audit, and scope

The deterministic archive contains 18 rational cases in dimensions one through six, 216 shell rows, six actual-support probes, three singular-face rows, and six boundary rows: 1653 audited leaves. Each case records the analytic exhaustion cutoff from the proof; the 12 displayed receipt shells do not prove (3). An independent checker reconstructs all shell multiplicities and values in 1930 assertions. A separate symbolic lane checks 36 identities, isolated replay demands two byte-identical outputs, and 72 repaired-hash or parser mutations must all be rejected.

Hostile tests alter the fastest set, exhaustion statement and cutoff, Morse/kernel dimensions, canonical rationals and decimals, exact JSON/YAML trees and types, and Route-A claims. They also attack duplicate keys, nonfinite tokens, YAML anchors, aliases, and merges. The three round variants are each rebuilt in two fresh directories at a fixed epoch and audited for warnings, text, pages, and fonts; the final PDF is a byte-identical alias of round 2.

**Collision and claim boundaries.** Earlier Fourier packages C206/C213/C217/C218/C261/C277 use different transport, hyperbolic, dispersive, fractional, or damping generators; C195 is nonlinear viscous Burgers. Here the theorem is the fourth-order conserved linear spinodal atlas in every finite dimension. It claims neither nonlinear Cahn–Hilliard saturation/coarsening nor literature novelty.

Under NO\_BAD\_EULER\_OR\_ROOT\_NUMBER, the Route-A tuple is

$$(A0\_FAIL, A1\_FAIL, A2\_FAIL, A3\_FAIL, A4\_FORMAL\_HINT).$$

Lattice shells are source Fourier geometry, not arithmetic local data (A0); stationary recurrence supplies no primitive-orbit bridge (A1); physical time is not a prime clock (A2); no target determinant or functional equation is constructed (A3). Self-adjointness is only a source-side formal hint, not a Hilbert–Pólya or target-zero result (A4). The overall verdict is ROUTE\_A\_REJECTED; Route B remains locked. No target Euler factor, root number, automorphy, divisor law, functional equation, or zero match is claimed.

**Reproducibility and AI use.** The release retains producer-independent validation, exact evidence, three substantive manuscript revisions, and a self-excluding manifest. A generative language model assisted with drafting and code scaffolding; independent executable lanes check formulas, artifacts, and scope, and final responsibility remains with the authors.

## Source lineage

Cahn and Hilliard introduced the nonuniform free-energy model, and Cahn’s spinodal paper is the historical owner token for the instability language. Elliott and Zheng provide classical analytical context. These references establish lineage only.

## References

- [1] J. W. Cahn and J. E. Hilliard, “Free Energy of a Nonuniform System. I. Interfacial Free Energy,” *J. Chem. Phys.* 28 (1958), 258–267. DOI: [10.1063/1.1744102](https://doi.org/10.1063/1.1744102).
- [2] J. W. Cahn, “On spinodal decomposition,” *Acta Metallurgica* 9 (1961), 795–801. DOI: [10.1016/0001-6160\(61\)90182-1](https://doi.org/10.1016/0001-6160(61)90182-1).
- [3] C. M. Elliott and S. Zheng, “On the Cahn–Hilliard equation,” *Arch. Rational Mech. Anal.* 96 (1986), 339–357. DOI: [10.1007/BF00251803](https://doi.org/10.1007/BF00251803).